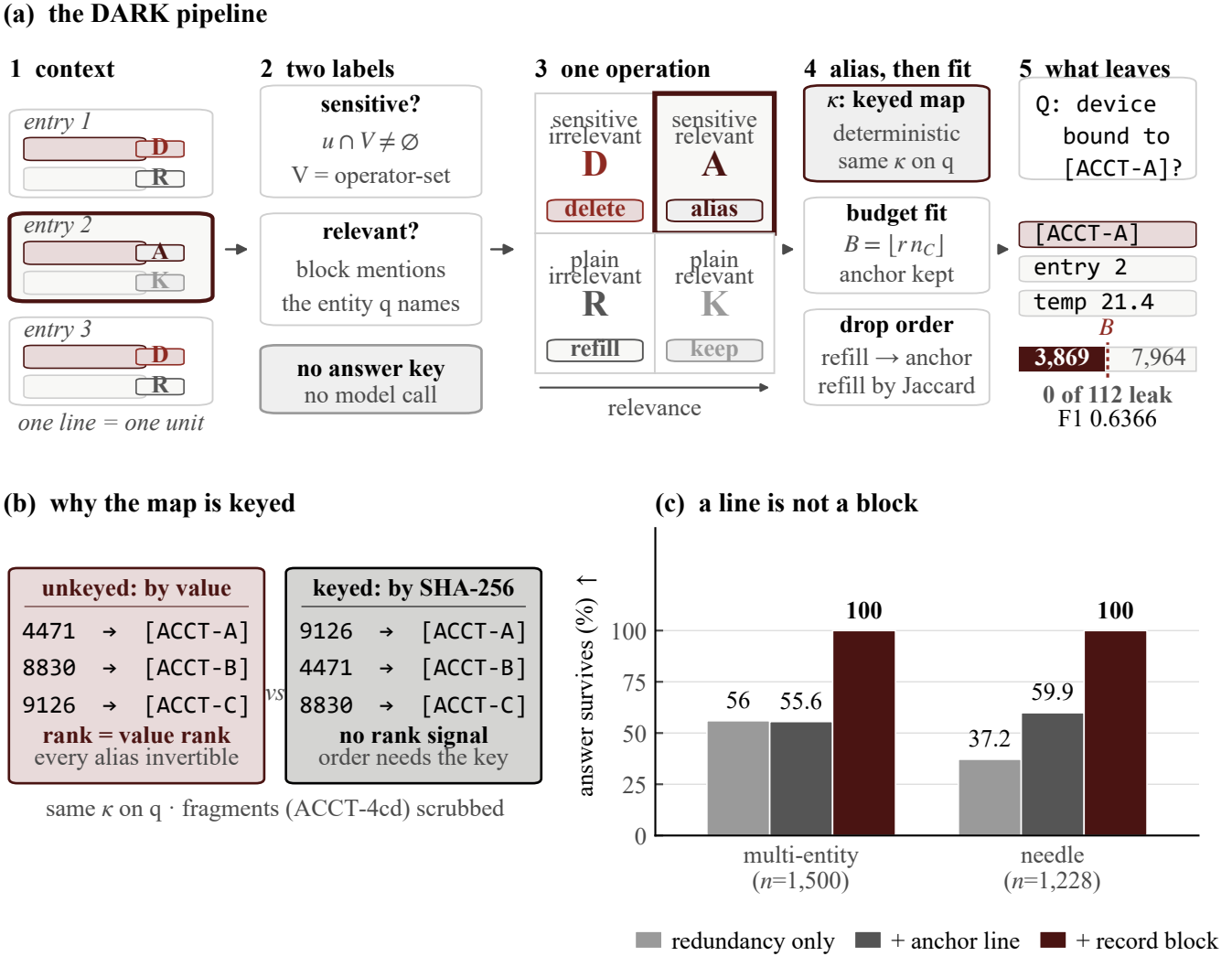

(a) the DARK pipeline
1 context
entry 1
D
R
entry 2
A
K
entry 3
D
R
one line = one unit
2 two labels
sensitive?
u ∩ V ≠ ∅
V = operator-set
relevant?
block mentions
the entity q names
no answer key
no model call
3 one operation
sensitive irrelevant
D
delete
sensitive relevant
A
alias
plain irrelevant
R
refill
plain relevant
K
keep
relevance
4 alias, then fit
κ: keyed map
deterministic
same κ on q
budget fit
B = ⌊r n_C⌋
anchor kept
drop order
refill → anchor
refill by Jaccard
5 what leaves
Q: device bound to [ACCT-A]?
[ACCT-A]
entry 2
temp 21.4
B
3,869
7,964
0 of 112 leak
F1 0.6366
(b) why the map is keyed
unkeyed: by value
4471 → [ACCT-A]
8830 → [ACCT-B]
9126 → [ACCT-C]
rank = value rank
every alias invertible
vs
keyed: by SHA-256
9126 → [ACCT-A]
4471 → [ACCT-B]
8830 → [ACCT-C]
no rank signal
order needs the key
same κ on q · fragments (ACCT-4cd) scrubbed
(c) a line is not a block
answer survives (%) ↑
100
75
50
25
0
56
55.6
100
37.2
59.9
100
multi-entity
(n=1,500)
needle
(n=1,228)
redundancy only
+ anchor line
+ record block